

To Begin with the Lab

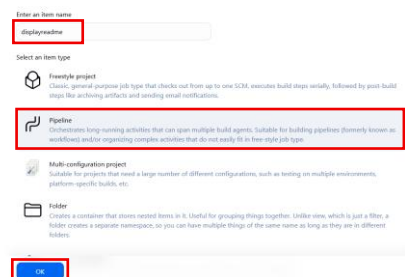
Job Setup for CloudFormation Parameters in Jenkinsfile

Summary of the Lab

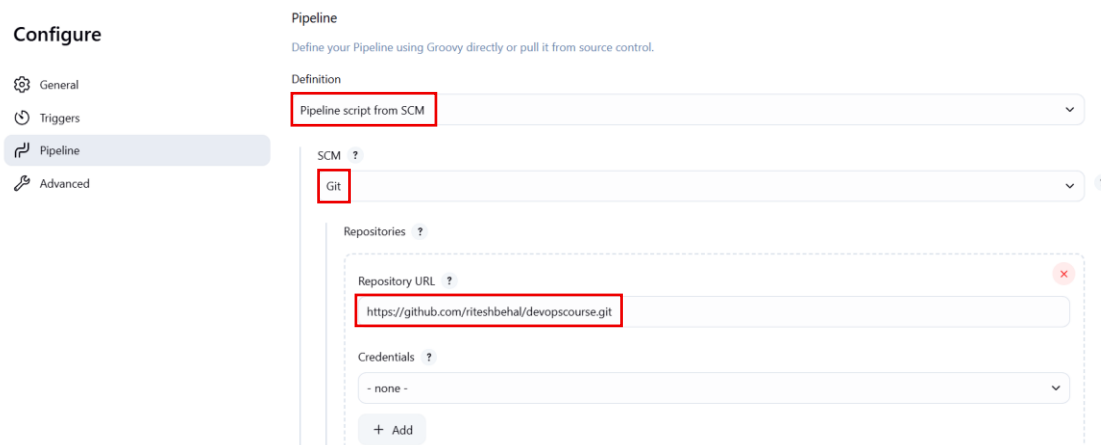
In this lab, we use files (01_s3cft.yml, 02_wordpress.json, 02_wordpress_Jenkinsfile, and Jenkinsfile) from the devopscourse GitHub repository to deploy infrastructure using Jenkins and AWS CloudFormation. A new Jenkins pipeline job is created using "Pipeline script from SCM," specifying the repository, branch (main), and correct script path. Parameters like region and file names are passed at build time. The Jenkinsfile executes a CloudFormation deployment. Later, 02_wordpress_Jenkinsfile is updated to hardcode parameters (e.g., key pair and region), committed, and rebuilt. AWS credentials and permissions must be configured. The stack creation results in EC2 instance provisioning.

Here we will use the files [01_s3cft.yml](#), [02_wordpress.json](#), [02_wordpress_Jenkinsfile](#) and [Jenkinsfile](#) from the root directory of the repository [devopscourse](#)

- Create a new Item.
- Give the name.
- Select the pipeline option.
- Click OK



- Under the pipeline, in definition, select the option Pipeline script from SCM.
- Under SCM, Select Git.
- Paste the URL of the Github Repository.



- Set the branch to main.

- Give the script path
- Note: give the path properly in the script path.
- In my case the jenkinsfile is under the folder pushdockerimage and inside the devopscourse repository.
- Click Save and click on build now.

Configure

- General
- Triggers
- Pipeline
- Advanced

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Add Branch

Repository browser ?

(Auto)

Additional Behaviours

Add

Script Path ?

02_wordpress_Jenkinsfile

☒ Lightweight checkout ?

Once the job is created, you will get the option ‘build with parameter’

Click again on ‘build with parameter’.

You will get options like this.

Give the names of the file properly and choose the current region that your instance is running on.

Click on build.

Pipeline exampleparameter

This build requires parameters:

STACK_NAME
Enter the CloudFormation Stack Name.

example-stack

PARAMETERS_FILE_NAME
Enter the Parameters File Name (Must contain file extension type *.properties)

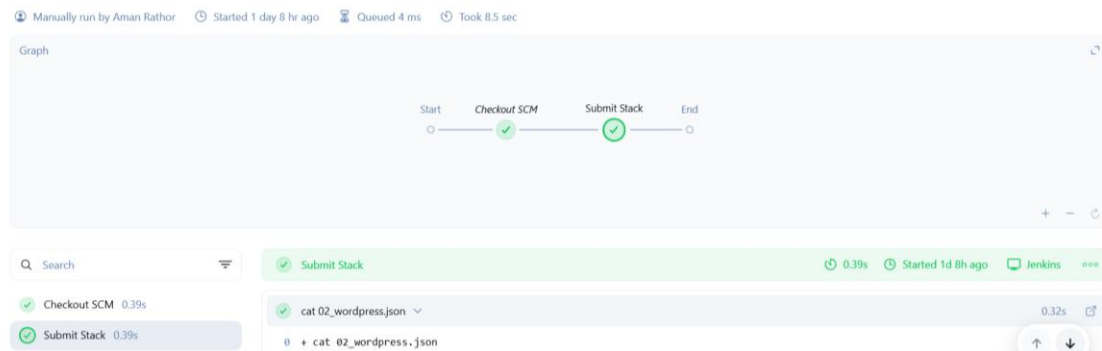
example-stack-parameters.properties

TEMPLATE_NAME
Enter the CloudFormation Template File Name

01_s3cft.yml

REGION
AWS Account Region

Build Cancel



Hardcoding CloudFormation Parameters in Jenkinsfile Demo

Summary of the Lab

We will continue with the same job that we have create previously.

Now we will make changes in the GitHub repository file named as '02_wordpress_Jenkinsfile'.

Comment the line number 21 and uncomment the line number 22

Don't forgot to change the region you are working in.

```

17 }
18 stages {
19   stage('Submit Stack') {
20     steps {
21       //sh "cat 02_wordpress.json"
22       sh "aws cloudformation deploy --stack-name ${STACK_NAME} --template-file ${TEMPLATE_NAME} --region 'us-west-2' --parameter-overrides
23         DBPassword=abcd1234 DBRootPassword=abcd1234 DBUser=admin KeyName=minikubetest"
24       //sh "aws cloudformation deploy --stack-name wordpress-test01 --template-file 02_wordpress.json --region 'us-west-2' --parameter-
25         overrides file://02_wordpress_parameter.properties"
26       //sh "aws cloudformation deploy --stack-name ${STACK_NAME} --template-file ${TEMPLATE_NAME} --region ${REGION} --parameter-overrides
27         file://${PARAMETERS_FILE_NAME}"
28     }
29   }
30 }

```

Use Control + Shift + ⌘ to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

```

stages {
  stage('Submit Stack') {
    steps {
      //sh "cat 02_wordpress.json"
      sh "aws cloudformation deploy --stack-name ${STACK_NAME} --template-file ${TEMPLATE_NAME} --region 'us-east-1' --parameter-overrides
        DBPassword=abcd1234 DBRootPassword=abcd1234 DBUser=admin KeyName=minikubetest"
      //sh "aws cloudformation deploy --stack-name wordpress-test01 --template-file 02_wordpress.json --region 'us-west-2' --parameter-
        overrides file://02_wordpress_parameter.properties"
      //sh "aws cloudformation deploy --stack-name ${STACK_NAME} --template-file ${TEMPLATE_NAME} --region ${REGION} --parameter-overrides
        file://${PARAMETERS_FILE_NAME}"
    }
  }
}

```

You can see that I have key pair existing with the name minikubetest

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Key pairs (9)

Find Key Pair by attribute or tag

	Name	Type	Created
☐	instancestore	rsa	2024/06/20 13:03 GMT+5:30
☐	ansible-key	rsa	2025/05/23 21:08 GMT+5:30
☐	4sd	rsa	2025/02/09 10:19 GMT+5:30
☐	north-virginia	rsa	2024/09/02 17:03 GMT+5:30
☐	mykey2	rsa	2025/07/17 07:38 GMT+5:30
☐	minikubetest	rsa	2025/07/08 12:43 GMT+5:30
☐	mykey	rsa	2025/07/06 12:36 GMT+5:30
☐	mykey1	rsa	2025/07/09 12:55 GMT+5:30
☐	KeyPair1	rsa	2025/07/14 12:15 GMT+5:30

Commit the changes and build the job again.

Here change the template name and choose the region you are in

Click build.

Note that you should configure the aws credential on the machine before building this job and also make sure that you should have the permission of CloudFormation.

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (6)

You can attach up to 10 managed policies.

Filter by Type

Search

All types

< 1 >

	Policy name	Type	Attached entities
☐	AdministratorAccess	AWS managed - job function	34
☐	AmazonEC2FullAccess	AWS managed	7
☐	AmazonS3FullAccess	AWS managed	31
☐	AmazonS3ReadOnlyAccess	AWS managed	17
☐	AWSCloudFormationFullAccess	AWS managed	1
☐	AWSLambdaRole	AWS managed	2

STACK_NAME

Enter the CloudFormation Stack Name.

example-stack

PARAMETERS_FILE_NAME

Enter the Parameters File Name (Must contain file extension type *.properties)

example-stack-parameters.properties

TEMPLATE_NAME

Enter the CloudFormation Template File Name

02_wordpress.json

REGION

AWS Account Region

us-east-1



Build

Cancel

It will create a stack under CloudFormation

The screenshot shows the AWS CloudFormation console's 'Stacks' page. On the left is a navigation menu with options like 'Stacks', 'StackSets', 'Exports', 'Infrastructure Composer', and 'Registry'. The main area displays a table of stacks. The 'example-stack' is highlighted with a red box. Its status is 'CREATE_IN_PROGRESS', created on 2025-07-18 at 18:01:53 UTC+0530. The description mentions it's an AWS CloudFormation Sample Template for WordPress.

Stack name	Status	Created time	Description
example-stack	CREATE_IN_PROGRESS	2025-07-18 18:01:53 UTC+0530	AWS CloudFormation Sample Template WordPress_Single_Instance: WordPress is web software you can use to create a beautiful website or blog. This template installs WordPress with a local MySQL database for storage. It demonstrates using the AWS CloudFormation bootstrap scripts to deploy WordPress. **WARNING** This template creates an Amazon EC2 instance. You will be billed for the AWS resources used if you create a stack from this template.
stack77	ROLLBACK_COMPLETE	2025-06-29 17:17:25 UTC+0530	Create S3 Bucket
awsstack29061	UPDATE_COMPLETE	2025-06-29 16:58:49 UTC+0530	Create S3 Bucket & SQS Queue

The screenshot shows the details of the 'example-stack' in the AWS CloudFormation console. The 'Stack info' tab is active, showing the stack ID, status, and description. The stack is in the 'CREATE_IN_PROGRESS' state. The description is the same as in the previous screenshot.

Stack ID
arn:aws:cloudformation:us-east-1:878893308172:stack/example-stack/2f684fe0-63d3-11f0-9e43-0affed541e0f

Status
CREATE_IN_PROGRESS

Detailed status
CONFIGURATION_COMPLETE

Status reason
Root stack

And this stack will create an instance.

Instances (3) Info

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

<

1

>

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public I
☐	kubernetes	i-067bbcf0e9bffd802	Running	t3.medium	3/3 checks passec	View alarms +	us-east-1c	ec2-54-
☐	Jenkins	i-0ce929966a84908f7	Running	t3.medium	3/3 checks passec	View alarms +	us-east-1c	ec2-3-8
☐		i-0571c2cc8559c8723	Running	t2.small	Initializing	View alarms +	us-east-1b	ec2-54-